Binary Number System

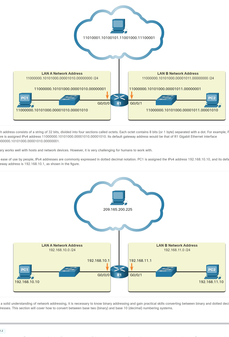
N1.0.1

Binary and IPv4 Addresses

IPv4 addresses begin as binary, a series of only 1s and 0s. These are difficult to manage, so network administrators must convert them to decimal. This topic shows you a few ways to do this.

Binary is a numbering system that consists of the digits 0 and 1 called bits. In contrast, the decimal numbering system consists of 10 digits consisting of the digits 0 – 9.

Binary is important for us to understand because hosts, servers, and network devices use binary addressing. Specifically, they use binary IPv4 addresses, as shown in the figure, to identify each other.



Each address consists of a string of 32 bits, divided into four sections called octets. Each octet contains 8 bits (or 1 byte) separated with a dot. For example, PC1 in the figure is assigned IPv4 address 11000000.10101000.00001010.00000010. Its default gateway address is the host at NIC B1 (Gigabit Ethernet interface 11000000.10101000.00001010.00000001).

Binary works well with hosts and network devices. However, it is very challenging for humans to work with.

For ease of use by people, IPv4 addresses are commonly expressed in dotted decimal notation. PC1's assigned the IPv4 address 192.168.10.10, and its default gateway address is 192.168.10.1, as shown in the figure.

For a solid understanding of network addressing, it is necessary to know binary addressing and gain practical skills converting between binary and dotted decimal IPv4 addresses. This section will cover how to convert between base two (binary) and base 10 (decimal) numbering systems.

N1.0.2

Video – Converting Between Binary and Decimal Numbering Systems

Click Play in the figure for a video demonstrating how to convert between binary and decimal numbering systems.

N1.0.3

Binary Positional Notation

Learning to convert binary to decimal requires an understanding of positional notation. Positional notation means that a digit represents different values depending on the "position" the digit occupies in the sequence of numbers. You already know the most common numbering system, the decimal (base 10) notation system.

The decimal positional notation system operates as described in the table.

Radix	10	10	10	10
Position in Number	3	2	1	0
Calculate	(10^3)	(10^2)	(10^1)	(10^0)
Position value	1000	100	10	1

The following bullets describe each row of the table:

- Row 1, Radix is the number base. Decimal notation is based on 10, therefore the radix is 10.
- Row 2, Position in number shows the positional value of each digit in a number. The digits are in base 10 (left, 0 [1st position], 1 [2nd position], 2 [3rd position], 3 [4th position]). These numbers also represent the exponential value (use to calculate the positional value in the 4th row).
- Row 3 calculates the positional value by taking the radix and raising it by the exponential value of its position in row 2.
- **Note** $10^0 = 1$
- Row 4 positional value represents units of thousands, tens, and ones.

To use the positional system, match a given number to its positional value. The example in the table illustrates how positional notation is used with the decimal number 1234.

	Thousands	Hundreds	Tens	Ones
Positional Value	1000	100	10	1
Decimal Number 1234C	1	2	3	4
Calculate	1 x 1000	2 x 100	3 x 10	4 x 1
Add them up...	1000	+ 200	+ 30	+ 4
Result	**1,234**			

In contrast, the binary positional notation operates as described in the table.

Radix	2	2	2	2	2	2	2	2
Position in Number	7	6	5	4	3	2	1	0
Calculate	(2^7)	(2^6)	(2^5)	(2^4)	(2^3)	(2^2)	(2^1)	(2^0)
Position value	128	64	32	16	8	4	2	1

The following bullets describe each row of the table:

- Row 1, Radix is the number base. Binary notation is based on 2, therefore the radix is 2.
- Row 2, Position in number shows the positional value of each digit in a number. The digits are in base 2 (left, 0 [1st position], 1 [2nd position], 2 [3rd position], 3 [4th position]). These numbers also represent the exponential value (use to calculate the positional value in the 4th row).
- Row 3 calculates the positional value by taking the radix and raising it by the exponential value of its position in row 2.
- **Note** $2^0 = 1$
- Row 4 positional value represents units of ones, twos, fours, eights, etc.

The example in the table below illustrates how a binary number 11000000 corresponds to the number 192. If the binary number had been 10101000, then the corresponding decimal number would be 168.

Positional Value	**128**	**64**	**32**	**16**	**8**	**4**	**2**	**1**
Binary Number 11000000	1	1	0	0	0	0	0	0
Calculate	1 x 128	1 x 64	0 x 32	0 x 16	0 x 8	0 x 4	0 x 2	0 x 1
Add Them Up...	128	+ 64	+ 0	+ 0	+ 0	+ 0	+ 0	+ 0
Result	**192**							

N1.0.4

Check Your Understanding – Binary Number System

Check your understanding of binary number systems by choosing the correct answer to the following questions.

1. Which is the binary equivalent to the 192.168.11.10 IP address?

- 11000000.11000000.00001011.00001010
- 11000000.10101000.00001011.00001010
- 11000000.10101000.00001010.00001011
- 11000000.10101000.00001011.00001010

2. Which of the following is the binary equivalent to the 172.16.31.30 IP address?

- 11000000.00010000.00011111.00011110
- 10101000.00010000.00011111.00011110
- 10101100.00010000.00011111.00011110
- 10101100.00010000.00011111.00011110

Check

Show Me

Reset

N1.0.5

Convert Binary to Decimal

To convert a binary IPv4 address to its dotted decimal equivalent, divide the IPv4 address into four 8-bit octets. Next apply the binary positional value to the first octet binary number and calculate accordingly.

For example, consider the 11000000.10101000.00001011.00001010 is the binary IPv4 address of a host. To convert the binary address to decimal, start with the first octet, as shown in the table. Enter the 8-bit binary number under the positional value of row 1 and then calculate to produce the decimal number 192. This number goes into the first octet of the dotted decimal notation.

Positional Value	**128**	**64**	**32**	**16**	**8**	**4**	**2**	**1**
Binary Number (11000000)	1	1	0	0	0	0	0	0
Calculate	128	64	32	16	8	4	2	1
Add Them Up...	128	+ 64	+ 0	+ 0	+ 0	+ 0	+ 0	+ 0
Result	**192**							

Next convert the second octet of 10101000 as shown in the table. The resulting decimal value is 168, and it goes into the second octet.

Positional Value	**128**	**64**	**32**	**16**	**8**	**4**	**2**	**1**
Binary Number (10101000)	1	0	1	0	1	0	0	0
Calculate	128	64	32	16	8	4	2	1
Add Them Up...	128	+ 0	+ 32	+ 0	+ 8	+ 0	+ 0	+ 0
Result	**168**							

Convert the third octet of 00001011 as shown in the table.

Positional Value	**128**	**64**	**32**	**16**	**8**	**4**	**2**	**1**
Binary Number (00001011)	0	0	0	0	1	0	1	1
Calculate	128	64	32	16	8	4	2	1
Add Them Up...	0	+ 0	+ 0	+ 0	+ 8	+ 0	+ 2	+ 1
Result	**11**							

Convert the fourth octet of 00001010 as shown in the table. This completes the IP address and produces **192.168.11.10**.

Positional Value	**128**	**64**	**32**	**16**	**8**	**4**	**2**	**1**
Binary Number (00001010)	0	0	0	0	1	0	1	0
Calculate	128	64	32	16	8	4	2	1
Add Them Up...	0	+ 0	+ 0	+ 0	+ 8	+ 0	+ 2	+ 0
Result	**10**							

N1.0.6

Activity – Binary to Decimal Conversions

Instructions

This activity allows you to practice 8-bit binary to decimal conversion as much as necessary. We recommend that you work with this tool until you are able to do the conversion without error. Convert the binary number shown in the table row to its decimal value.

Enter decimal answer below.

Decimal Value								
Base	2	2	2	2	2	2	2	2
Exponent	7	6	5	4	3	2	1	0
Position	128	64	32	16	8	4	2	1
00	1	0	0	1	1	0	1	1

Binary Number

Check | New Number | Show Me | Reset

N1.0.7

Decimal to Binary Conversion

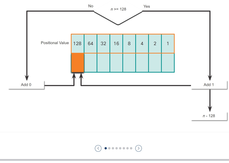
It is also necessary to understand how to convert a dotted decimal IPv4 address to binary. A useful tool is the binary positional value table.

Click each position starting at 128 and work your way from left to right to the 1 position.

128	64	32	16	8	4	2	1

Is the decimal number of the octet (x) equal to or greater than the most significant bit (128)?

- If no, then enter binary **0** in the **128** positional value
- If yes, then add a binary 1 in the **128** positional value and subtract **128** from the decimal number.



N1.0.8

Decimal to Binary Conversion Example

To help understand the process, consider the IP address **192.168.10.11**.

The first octet number **192** is converted to binary using the previously explained positional notation process.

It is possible to bypass the process of subtraction with easier or smaller decimal numbers. For instance, notice that it is fairly easy to calculate the third octet converted to a binary number without actually going through the subtraction process (8 + 2 = 10). The binary value of the third octet is **00001010**.

The fourth octet is 11 (8 + 2 + 1). The binary value of the fourth octet is **00001011**.

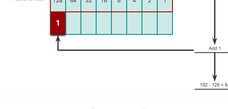
Converting between binary and decimal may seem challenging at first, but with practice it should become easier over time.

Click each step to see the conversion of the IP address of **192.168.10.11** into binary.

Step 1 | Step 2 | Step 3 | Step 4 | Step 5 | Step 6 | Step 7 | Step 8 | Step 9 | Step 10 | Step 11

In the first octet number **192** equal to or greater than the high order bit **128**?

- Yes it is, therefore add a 1 to the high order positional value to represent **128**.
- Subtract **128** from **192** to produce a remainder of **64**.



N1.0.9

Activity – Decimal to Binary Conversions

Instructions

This activity allows you to practice decimal conversions to 8-bit binary values. We recommend that you work with this tool until you are able to do the conversion without error. Convert the decimal number shown in the Decimal Value row to its binary bits.

Decimal Value					192			
Base	2	2	2	2	2	2	2	2
Exponent	7	6	5	4	3	2	1	0
Position	128	64	32	16	8	4	2	1
00								

Check | New Number | Show Me | Reset

N1.0.10

Activity – Binary Game

This is a fun way to learn binary numbers for first octet.

Game Link: https://learningnetwork.cisco.com/docs/DOC-1803

You will need to log in to cisco.com to use this link. It will be necessary to create an account if you do not already have one.

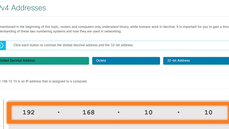
There are also a variety of free mobile binary games. Search for "binary game" in your app store.

N1.0.11

IPv4 Addresses

As mentioned in the beginning of this topic, routers and computers only understand binary, while humans work in decimal. It is important for you to gain a thorough understanding of these two numbering systems and how they are used in networking.

Click each button to contrast the dotted decimal address and the 32-bit address.

Dotted Decimal Address | Octets | 32-bit Address

192.168.10.10 is an IP address that is assigned to a computer.